

Pandas Beginner → Interview Practice

A practical coding workbook created by ChatGPT

Use this PDF to practice Pandas step by step. Try each problem yourself before checking the solution.

Core Code Practice

1. Import Pandas

```
import pandas as pd
print(pd.__version__)
```

2. Create a Series

```
s = pd.Series([10, 20, 30, 40])
print(s)
```

3. Create a DataFrame

```
data = {'Name': ['Asha', 'Rahul', 'John'], 'Age': [22, 25, 21]}
df = pd.DataFrame(data)
print(df)
```

4. View first rows

```
print(df.head())
print(df.head(2))
```

5. View last rows

```
print(df.tail())
```

6. Check shape and columns

```
print(df.shape)
print(df.columns)
```

7. Select one column

```
print(df['Age'])
```

8. Select multiple columns

```
print(df[['Name', 'Age']])
```

9. Filter rows

```
print(df[df['Age'] > 21])
```

10. Add a new column

```
df['City'] = ['Kochi', 'Kannur', 'Kozhikode']
print(df)
```

11. Rename a column

```
df = df.rename(columns={'Age': 'Years'})
print(df)
```

12. Sort values

```
print(df.sort_values('Years'))
```

13. Check data information

```
print(df.info())
```

14. Descriptive statistics

```
print(df.describe())
```

15. Find missing values

```
print(df.isnull().sum())
```

16. Fill missing values

```
df['Years'] = df['Years'].fillna(df['Years'].mean())
```

17. Drop missing rows

```
df = df.dropna()
```

18. Remove duplicate rows

```
df = df.drop_duplicates()
```

19. Value counts

```
print(df['City'].value_counts())
```

20. GroupBy

```
sales = pd.DataFrame({'City':['Kochi','Kochi','Kannur'], 'Sales':[100,200,150]})  
print(sales.groupby('City')['Sales'].sum())
```

21. Read a CSV file

```
df = pd.read_csv('data.csv')  
print(df.head())
```

22. Save a DataFrame

```
df.to_csv('output.csv', index=False)
```

23. Apply a condition with multiple filters

```
result = df[(df['Age'] >= 20) & (df['Age'] <= 30)]  
print(result)
```

24. Merge two DataFrames

```
result = pd.merge(df1, df2, on='customer_id', how='inner')  
print(result)
```

25. Concatenate DataFrames

```
result = pd.concat([df1, df2], ignore_index=True)  
print(result)
```

Interview Practice Questions

1. What is Pandas?
2. What is a Series?
3. What is a DataFrame?
4. What is the difference between Series and DataFrame?
5. What does head() do?
6. What does tail() do?
7. What does shape return?
8. What is the use of info()?
9. What is the use of describe()?
10. How do you read a CSV file?
11. How do you handle missing values?
12. Difference between dropna() and fillna().
13. How do you remove duplicate rows?
14. How do you filter rows using a condition?
15. What is groupby() used for?
16. What is the difference between merge() and concat()?
17. How do you sort a DataFrame?
18. What does value_counts() do?
19. How do you select multiple columns?
20. Why is Pandas important in Data Science?

Mini Practice Dataset

```
data = {
    'Name': ['Asha', 'Rahul', 'John', 'Meera', 'Arun'],
    'Age': [22, 25, 21, 28, 24],
    'City': ['Kochi', 'Kannur', 'Kochi', 'Calicut', 'Kannur'],
    'Salary': [25000, 32000, 28000, 40000, 30000]
}
df = pd.DataFrame(data)
print(df)
```

Practice these tasks without looking at the answers:

1. Display the first 3 rows.
2. Display only Name and Salary.
3. Find people with Salary greater than 30000.
4. Find the average Salary.
5. Find the maximum Age.
6. Count employees by City.
7. Sort employees by Salary from highest to lowest.
8. Add a new column called Experience.
9. Save the result as employees.csv.
10. Group by City and calculate average Salary.

Reference

This workbook is an original practice resource created by ChatGPT. For current Pandas concepts and API details, use the official Pandas documentation.